

Climate State and Precession Phase Organize DO-Like Asian Monsoon Transition

Abstract

Millennial-scale climate variability is a prominent feature of glacial climates, yet how slow climate state and orbital phase jointly organize the timing of individual abrupt transitions remains unclear. Here, we test this question using 196 objectively identified weak- and strong-monsoon starts from the Chinese speleothem composite over the past 640 kyr. Rayleigh tests show that event timing exhibits opposite preferences in precession phase, with strong- and weak-monsoon starts clustering near precession minima and maxima, respectively. A history-conditioned predictive-information framework shows that LR04 benthic $\delta^{18}\text{O}$ and atmospheric CO_2 significantly improve event-rate prediction after same-type event history and composite sampling resolution are controlled. Precession phase then adds a comparable layer of information after this glacial background state is included. Lagged diagnostics further suggest that LR04 and CO_2 behave as slower background-state predictors, whereas precession phase contributes near-contemporaneous orbital-seasonal organization.

Introduction

Millennial-scale climate variability (MCV) is a major expression of abrupt climate change in glacial climate systems. Its canonical manifestation is the Dansgaard–Oeschger (DO) sequence in Greenland ice cores, where rapid stadial–interstadial transitions record major reorganizations of the North Atlantic climate system (*Rasmussen et al.*, 2014). Yet MCV was not confined to Greenland. Marine, ice-core, and speleothem archives show that abrupt millennial variability occurred across multiple Pleistocene glacial cycles and was expressed in both high-latitude and low-latitude monsoon systems (*Barker et al.*, 2011; *Sun et al.*, 2021; *Hodell et al.*, 2023). Speleothem records provide a particularly powerful archive because their U–Th chronologies allow abrupt hydroclimate changes to be compared directly with ice-core event sequences. During the last glacial period, independently dated speleothem records indicate that abrupt Greenland warmings were accompanied by near-synchronous hydroclimate shifts in monsoon and mid-latitude regions (*Corrick et al.*, 2020). Such speleothem transitions can therefore be treated as precisely dated DO-like Asian monsoon expressions of MCV.

The intensity and occurrence of MCV were strongly conditioned by the slowly evolving background climate state. Proxy syntheses and model experiments indicate that abrupt millennial-scale variability was favored under intermediate glacial conditions, whereas interglacial and fully glacial end-member states were comparatively unfavorable for MCV (*Dansgaard et al.*, 1993; *Barker et al.*, 2011; *Kawamura et al.*, 2007; *Malmierca-Vallet and Sime*, 2023). Changes in ice volume, ice-sheet geometry, greenhouse forcing, sea ice, and ocean circulation can shift the climate system between stable and oscillatory regimes (*Zhang et al.*, 2017; *Vettoretti et al.*, 2022; *Zhang et al.*,

2021; *Malmierca-Vallet and Sime, 2023*). Atmospheric CO₂ (*Bereiter et al., 2015*) provides a complementary control because intermediate CO₂ levels can place the AMOC close to a threshold or bistable window (*Zhang et al., 2017; Vettoretti et al., 2022*).

Orbital forcing further modulated MCV amplitude within this state-dependent window, but not through a single pacing mechanism. Long marine and terrestrial records show persistent obliquity- and precession-band structure in the millennial-variance envelope, with obliquity dominating early Pleistocene and MPT modulation, and precession-band and 100-kyr power strengthening after the growth of larger late-Pleistocene ice sheets (*Sun et al., 2021; Hodell et al., 2023; Thirumalai et al., 2020*). These orbital imprints likely operated through different pathways: obliquity via ice volume, sea ice, and AMOC stability, and precession via seasonal insolation, tropical hydroclimate, and monsoon-ITCZ shifts (*Sun et al., 2021; Zhang et al., 2021*).

Amplitude modulation, however, does not necessarily imply event pacing. A larger millennial-scale envelope under a particular orbital configuration may reflect the amplification of internally generated transitions, rather than external control on when individual events occur. In stochastic or excitable climate systems, abrupt regime shifts can arise when internal variability crosses a threshold, producing clustered or intermittent events without direct orbital pacing (*Ditlevsen et al., 2005; Kleppin et al., 2015*). Conversely, climate-model experiments suggest that orbital configuration can alter DO-like oscillatory regimes by changing tropical hydroclimate, high-latitude sea ice, and ocean circulation, thereby switching millennial-scale AMOC variability on or off under otherwise comparable boundary conditions (*Zhang et al., 2021; Malmierca-Vallet and Sime, 2023*). The key unresolved proxy question is therefore not only whether orbital forcing modulates MCV amplitude, but whether it also organizes the timing of individual stadial-interstadial-like transitions.

Testing this event-timing question requires a long, precisely dated sequence of abrupt transitions. The 640 kyr Chinese speleothem composite of *Cheng et al. (2016)*, together with the objectively identified weak- and strong-monsoon starts of *Rousseau et al. (2023)*, provides a rare opportunity to examine event timing over multiple orbital cycles. These weak and strong starts are not Greenland-defined DO events themselves, but DO-like Asian monsoon transitions that record the monsoon-sector expression of MCV (*Cheng et al., 2016*).

Here we test how slow climate state and orbital phase jointly organize the timing of these DO-like Asian monsoon transitions. Using the 0–640 kyr BP weak- and strong-monsoon start catalogues of *Rousseau et al. (2023)*, we ask three linked questions. First, do weak- and strong-monsoon starts cluster at preferred precession or obliquity phases? Second, do slow background-state predictors represented by LR04 ice volume and atmospheric CO₂ improve event-rate prediction after event history and sampling resolution are controlled? Third, does precession phase add information beyond that background state? By shifting the focus from variance amplitude to transition probability, this event-based framework tests whether glacial background conditions and orbital configuration together helped organize abrupt monsoon-state changes. The data and phase framework are summarized in Figure 1.

Methods

We analyzed the complete 0.4–4 kyr KS detection-window weak- and strong-monsoon start chronologies identified by *Rousseau et al. (2023)* from the Chinese speleothem composite of *Cheng et al. (2016)*. This catalogue contains 196 events: 96 strong monsoon starts and 100 weak monsoon starts. All tests were applied independently to the two event catalogues. The predictive-information models were fitted on the common support for the event and covariate series, after applying the history-window requirement described below, leaving 95 strong starts and 98 weak starts.

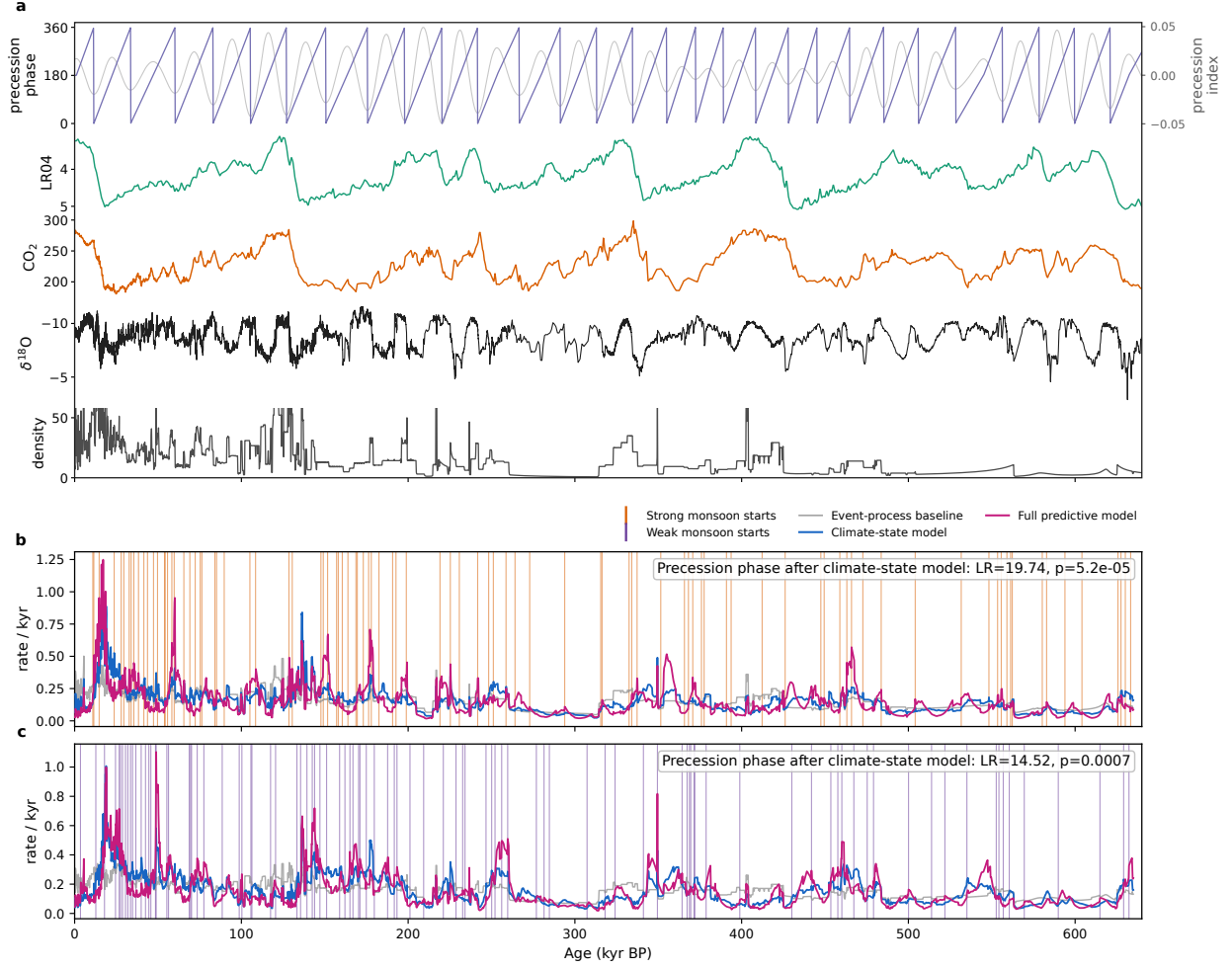


Figure 1: Data and phase framework. (a) Input series used in the predictive-information model, including precession phase and raw precession index (*Laskar et al., 2004*), LR04 benthic $\delta^{18}\text{O}$ (*Lisiecki and Raymo, 2005*), atmospheric CO_2 (*Bereiter et al., 2015*), the Chinese speleothem $\delta^{18}\text{O}$ composite (*Cheng et al., 2016*), and composite sampling density. (b) Fitted event rates for strong monsoon starts from the event-process baseline, the climate-state model, and the full predictive model. Vertical lines mark observed strong-start event bins. (c) Same as (b), but for weak monsoon starts.

Before fitting the predictive models, we used circular Rayleigh tests to ask whether event phases are uniformly distributed with respect to precession and obliquity. These tests provide diagnostic context but do not adjust for event history, sampling resolution, or glacial background state; details are given in Text S1, and the precession-phase diagnostics are summarized in Figure 2.

The main analysis used a history-conditioned binned Poisson predictive-information framework. The 0–640 kyr BP interval was divided into bins of width $\Delta t = 0.2$ kyr. For bin i , the response was the observed event count Y_i , modeled conditionally on the older event history and external covariates as

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \Delta t_i \lambda_i,$$

where λ_i is the event rate per kyr. The log link was applied to the rate,

$$\log \lambda_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta},$$

or equivalently

$$\log \mu_i = \log \Delta t_i + \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta},$$

so $\log \Delta t_i$ entered as an offset. The fitted log-likelihood was

$$\ell(\boldsymbol{\beta}) = \sum_i [Y_i \log \mu_i - \mu_i - \log(Y_i!)] .$$

To reduce the risk of attributing catalogue structure to external forcing, every main predictive model was built on an event-process baseline containing two nuisance controls. The first was same-type event history,

$$H_i = \sum_{t_i < t_j \leq t_i + 5 \text{ kyr}} Y_j,$$

the number of events of the same type in the immediately older 5 kyr interval. Age increases into the past, so H_i is a predictable history covariate: the likelihood is interpreted as a product of conditional bin probabilities rather than as a Poisson model with independent event histories. The second nuisance control was local sampling resolution of the Cheng et al. composite. For each composite $\delta^{18}\text{O}$ age point, local spacing was defined as the median of the adjacent younger and older age gaps; this spacing series was interpolated to event-bin centers, log-transformed, and range-scaled like the other scalar predictors. This resolution term accounts, in a first-order way, for the possibility that apparent event density covaries with data spacing.

The core model sequence was therefore nested as follows. The event-process baseline included H_i and sampling resolution. The climate-state model added LR04 benthic $\delta^{18}\text{O}$ and atmospheric CO_2 . The full predictive model further added precession phase:

$$\log \lambda_i = \beta_0 + \beta_H H_i + \beta_R R_i + \beta_1 \text{LR04}_i + \beta_2 \text{CO}_{2,i} + \beta_s \sin \theta_i + \beta_c \cos \theta_i.$$

Here R_i is the scaled log-resolution term and θ_i is precession phase at the bin center. Precession phase was defined from the precession-index waveform (*Laskar et al.*, 2004) by assigning phase 0 to local minima and phase π to local maxima, with linear interpolation of unwrapped phase between adjacent extrema. LR04, CO_2 , and scalar sensitivity predictors were interpolated to bin centers and range-scaled to zero mean and unit observed range. Phase was represented by $\sin \theta_i$ and $\cos \theta_i$ because phase is circular.

Models were fitted by maximum likelihood using the L-BFGS-B method through SciPy’s optimize.minimize routine (*Virtanen et al.*, 2020); the stationary model has the analytic maximum-likelihood rate N/T . Models were ranked using sample-size-corrected Akaike information criterion (AICc). Nested models were compared with likelihood-ratio tests,

$$\text{LR} = 2(\ell_{\text{full}} - \ell_{\text{reduced}}),$$

using a chi-square reference distribution with degrees of freedom equal to the number of added coefficients. We also report the same log-likelihood gain as model-based predictive information,

$$I_{\text{bits/event}} = \frac{\ell_{\text{full}} - \ell_{\text{reduced}}}{\ln 2 N_{\text{events}}}.$$

The two central comparisons were LR04+CO₂ after the event-process baseline and precession phase after the climate-state model. The latter comparison used the climate-state model as the reduced model and the full predictive model as the full model. Because adjacent bins share strongly overlapping history windows and events are sparse, the asymptotic χ^2 reference for LR tests could be optimistic in finite samples. Therefore, for these two central nested comparisons, we also used reduced-model parametric bootstrap tests that regenerated event catalogues, recomputed same-type history, and refitted the reduced and full models to obtain an empirical LR null (Text S5).

For models containing $\sin \theta$ and $\cos \theta$, the preferred phase and phase-response amplitude were derived from the two coefficients. If the phase component is

$$\beta_s \sin \theta + \beta_c \cos \theta,$$

then the phase of maximum fitted rate is

$$\theta^* = \text{atan2}(\beta_s, \beta_c) \bmod 2\pi,$$

and the amplitude is

$$A = (\beta_s^2 + \beta_c^2)^{1/2}.$$

The fitted maximum-to-minimum rate ratio over one precession cycle is $\exp(2A)$.

We used the same nested-Poisson logic as a lagged predictive-information diagnostic. This analysis was intended to compare temporal alignment among predictors. For each driver X and lag τ , we compared the event-process baseline with a full model adding $X(t_i + \tau)$. Positive lag means that the driver was sampled from an older climate state relative to the event bin. Lags from 0 to 20 kyr were scanned in 0.2 kyr steps. We ran three scans: each driver separately after the event-process baseline; precession phase after LR04 and CO₂ sampled at the same lag; and precession phase after LR04 and CO₂ fixed at their own best single-driver lags. Benjamini–Hochberg false-discovery-rate q values were computed across scanned lags within each dataset and driver family.

We performed several sensitivity experiments. Additional-forcing tests added Antarctic temperature, obliquity, eccentricity, and 65°N summer-solstice insolation to the full predictive model for sensitivity purposes. We tested each extra forcing individually, all extras jointly, and drop-one contributions within the extended model. Bin-width sensitivity repeated the full predictive-model comparison for bin widths of 0.2, 0.4, 0.6, 0.8, and 1.0 kyr. KS-window sensitivity repeated the core tests for the alternative 0.6–4 kyr Rousseau transition catalogue. Composite-age sensitivity generated 500 rank-preserving age-randomized catalogues using an age-uncertainty envelope increasing from 1 kyr at 0 kyr BP to 7.4 kyr at 640 kyr BP—this envelope intentionally adopts conservative values that likely exceed the true composite age uncertainty, because the underlying U–Th dates of individual speleothem records have two-sigma errors on the order of a few hundred years in the youngest portion (*Cheng et al.*, 2016)—and repeated the Rayleigh and predictive-information comparisons for each realization. Finally, as an independent event-catalogue check, we repeated the Rayleigh and predictive-information analyses for the *Barker et al.* (2011) variable-threshold D–O warming catalogue, treating these events as strong monsoon start analogues; details are given in Text S6.

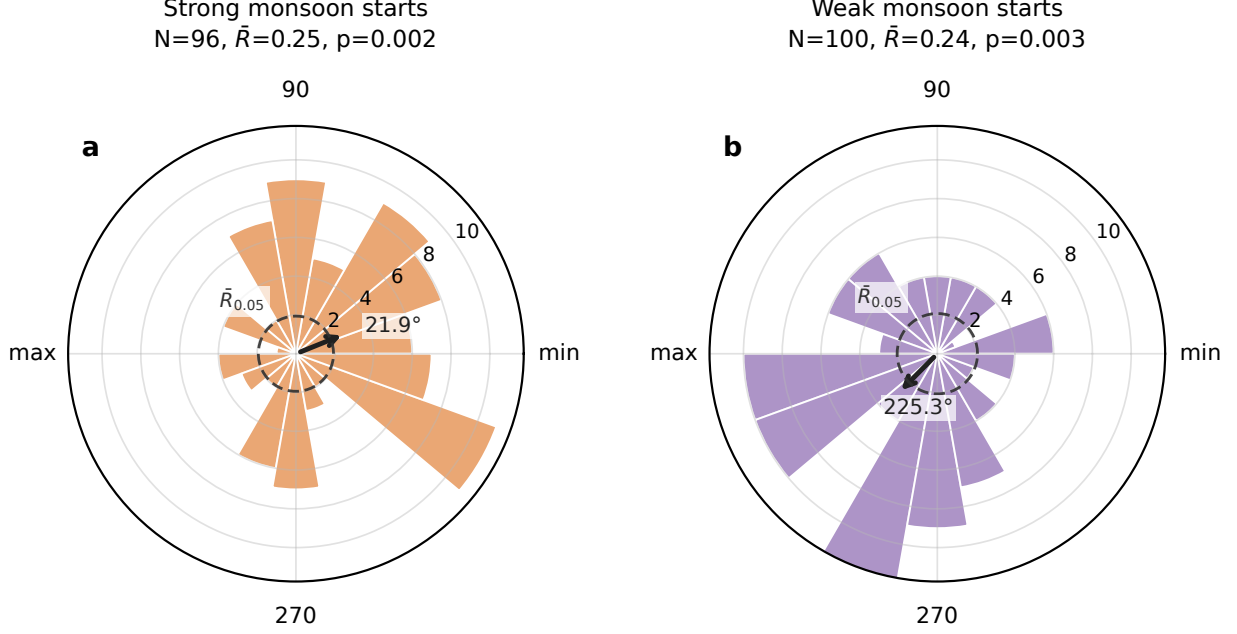


Figure 2: Precession-phase clustering of monsoon starts. (a,b) Rayleigh tests for clustering of strong and weak monsoon starts in precession phase; arrows show the mean resultant direction and dashed circles mark the $p = 0.05$ threshold. Precession index from *Laskar et al. (2004)*; event catalogue from *Rousseau et al. (2023)*.

Results

The supporting Rayleigh tests first identify precession as the relevant orbital phase variable. Strong and weak monsoon starts show significant precession-phase clustering, with mean phases of 21.9° and 225.3° , respectively (Figure 2). In contrast, obliquity phases are not significantly clustered for either event type (Figure S1).

The predictive-information models show contributions from event-process structure, slow background state, and precession phase. Relative to a stationary rate, the event-process baseline improved the likelihood for strong starts ($\text{LR} = 16.61$, $p = 2.48 \times 10^{-4}$) and weak starts ($\text{LR} = 12.27$, $p = 0.00216$). Adding $\text{LR04} + \text{CO}_2$ to the event-process baseline further improved prediction for strong starts ($\text{LR} = 13.73$, $p = 0.00104$, $\Delta\text{AICc} = -9.72$) and weak starts ($\text{LR} = 21.69$, $p = 1.95 \times 10^{-5}$, $\Delta\text{AICc} = -17.68$). In information units, these gains are 0.104 and 0.160 bits/event, equivalent to roughly 7% and 12% higher average per-event likelihood for the observed strong- and weak-start sequences, respectively. Precession phase supplied an additional, comparable layer of predictive information after the climate-state model was included. For strong starts, the likelihood-ratio statistic was $\text{LR} = 19.74$ ($p = 5.16 \times 10^{-5}$, $\Delta\text{AICc} = -15.73$, 0.150 bits/event). For weak starts, the corresponding statistic was $\text{LR} = 14.52$ ($p = 7.03 \times 10^{-4}$, $\Delta\text{AICc} = -10.50$, 0.107 bits/event). These information gains correspond to an approximately 11% and 8% increase in the average per-event likelihood assigned to the observed strong- and weak-start sequences, respectively. Reduced-model bootstrap tests gave larger but still significant empirical p values for both $\text{LR04} + \text{CO}_2$ after the event-process baseline and precession phase after the climate-state model ($p_{\text{boot}} < 0.003$; Table S1). A companion visualization without the event-process baseline shows the slower $\text{LR04} + \text{CO}_2$ rate modulation and the additional precession-phase structure more directly

(Figure S3).

The fitted phase preferences agree with the direct Rayleigh tests and show opposite sectors for the two transition directions (Figure 2). In the event-process-baseline plus precession-phase model, strong starts prefer 20.7° , close to the Rayleigh mean of 21.9° . In the full predictive model, the preferred phase shifts to 2.1° , and the maximum-to-minimum rate ratio over the precession cycle is 4.02. This means that, with the other fitted predictors held fixed, the modeled strong-start hazard is about four times higher at the preferred precession phase than at the opposite phase. Weak starts prefer 225.1° in the event-process-baseline plus precession-phase model and 232.8° in the full predictive model, with a full-model rate ratio of 3.21. Thus, precession phase contributes information after conditioning on both event-process structure and slow glacial background, and the inferred direction of the phase effect is consistent across independent tests.

The full predictive model is the main scientific reference model, but the AICc rankings also indicate that LR04 and CO₂ partly overlap (Figure 3), consistent with the high correlation between them (Figure S4). For strong starts, the lowest AICc model contains LR04 and precession phase but not CO₂, and the full predictive model ranks second. For weak starts, the lowest AICc model contains CO₂ and precession phase but not LR04, again with the full predictive model ranking second. Drop-one tests show the same asymmetry: after CO₂+precession phase, LR04 is not significant for weak starts, whereas after LR04+precession phase, CO₂ remains significant for weak starts ($p = 0.0118$) (Figure 3). These results should not be read as a clean separation of ice volume and greenhouse forcing. Rather, LR04 and CO₂ jointly capture slow glacial background structure, while precession phase remains the clearest orbital timing term after that structure is included.

The additional-forcing sensitivity tests do not identify another independent predictor comparable to precession phase (Figure S5). After the extended baseline model, defined here as the full predictive model before adding extra forcings, was fitted, adding Antarctic temperature, obliquity, eccentricity, or 65°N summer-solstice insolation one at a time did not significantly improve the strong-start model ($p = 0.452\text{--}0.953$) or the weak-start model ($p = 0.127\text{--}0.877$) (Figure S5). Adding all four extra forcings jointly also failed to improve prediction for strong starts ($p = 0.932$, $\Delta\text{AICc} = +7.20$) or weak starts ($p = 0.455$, $\Delta\text{AICc} = +4.40$) (Figure S5). The only suggestive but non-significant case is 65°N insolation for weak starts, which slightly lowers AICc when added alone ($\Delta\text{AICc} = -0.32$) and in the drop-one test ($\Delta\text{AICc} = -1.20$), but has likelihood-ratio $p = 0.127$ and $p = 0.073$, respectively (Figure S5). We therefore treat this as weak sensitivity, not as evidence for an additional independent forcing (Figure S6).

The bin-width, KS-window, and age-uncertainty sensitivity experiments support the robustness of the precession-phase result. Across bin widths from 0.2 to 1.0 kyr, precession phase after the climate-state model remains significant for both event catalogues. Strong-start preferred phases stay between 2.1° and 5.5° , and weak-start preferred phases stay between 232.8° and 234.5° ; all phase shifts relative to the 0.2 kyr baseline are less than 3.4° . Using the narrower 0.6–4 kyr KS-window catalogue also preserves Rayleigh precession clustering and the precession-phase likelihood-ratio result after the climate-state model for both strong and weak starts (Table S2). Finally, in 500 age-randomized catalogues, the precession-phase term after the climate-state model remains significant in 100% of realizations for both event types, and Rayleigh precession clustering remains significant in 99.6% of strong-start and 99.0% of weak-start age-randomized catalogues.

The independent *Barker et al.* (2011) D-O warming catalogue gives a weaker but useful check because it is not derived from the Cheng speleothem composite or the *Rousseau et al.* (2023) KS transition-detection workflow. Direct Rayleigh tests show only weak precession-phase clustering for the Barker events ($\bar{R} = 0.118\text{--}0.189$, $p = 0.081\text{--}0.177$), so this catalogue does not reproduce the stronger circular clustering seen in the Rousseau speleothem detections. In the predictive-information framework, however, the Barker catalogue gives a similar two-step result to the main

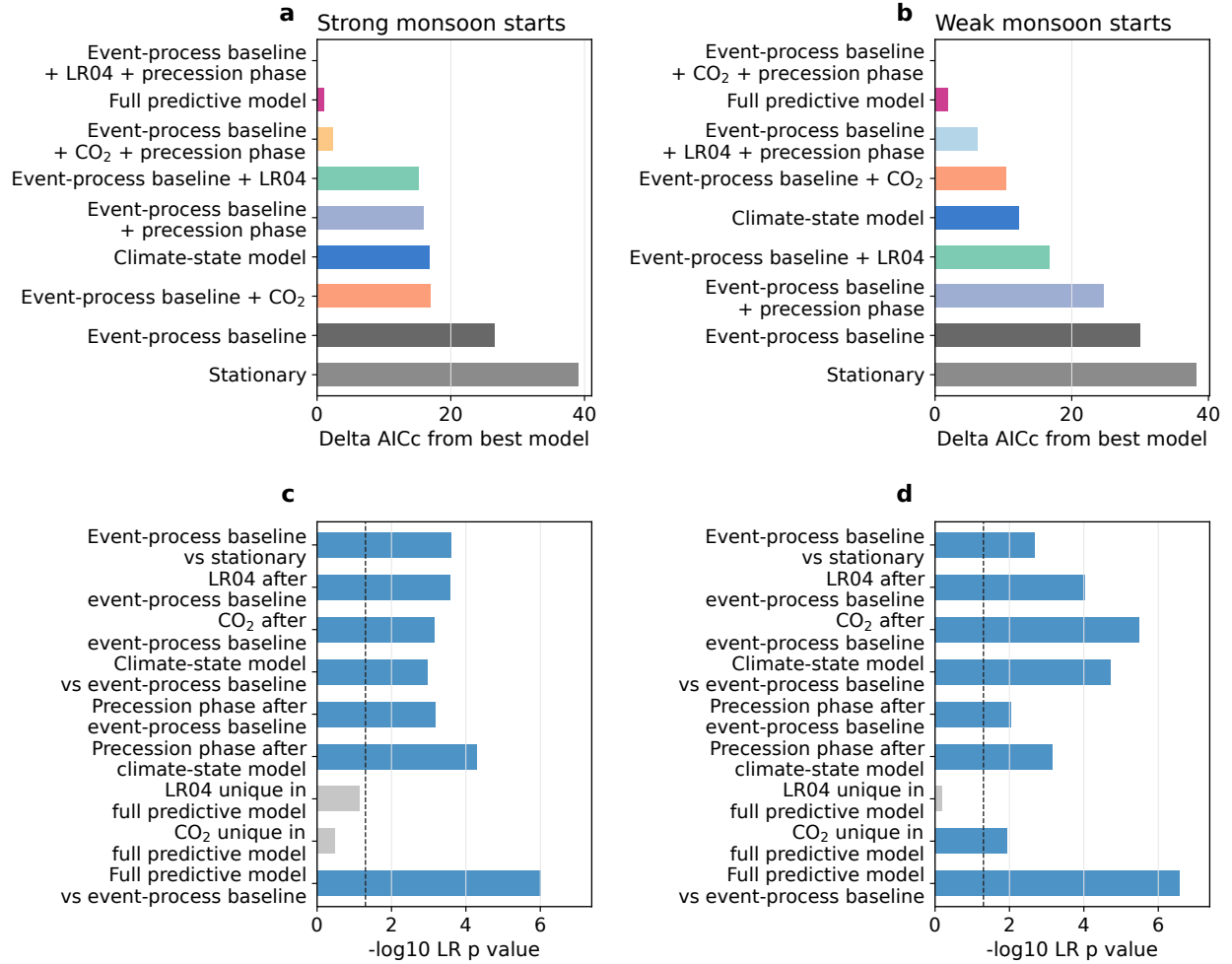


Figure 3: Predictive-information model comparison. Model comparison results are shown for (a) strong and (b) weak monsoon starts. Bars show ΔAICc relative to the lowest-AICc model within each event catalogue. The event-process baseline contains same-type event history and Cheng composite sampling resolution (*Cheng et al.*, 2016); the climate-state model further includes LR04 (*Lisiecki and Raymo*, 2005) and CO₂ (*Bereiter et al.*, 2015); and the full predictive model adds precession phase (*Laskar et al.*, 2004). (c,d) Nested likelihood-ratio tests for the same event catalogues. Bars show $-\log_{10}(p)$, and the dashed line marks $p = 0.05$; bars to the right of this line are significant.

analysis: LR04+CO₂ improves prediction after the EDC event-process baseline ($\Delta\text{AICc} = -7.31$ to -4.32 , $p = 0.00348$ – 0.0155 , 0.058 – 0.093 bits/event), and precession phase adds further information after the Barker climate-state model ($\Delta\text{AICc} = -3.94$ to -2.78 , $p = 0.0188$ – 0.0334 , 0.039 – 0.081 bits/event; Text S6 and Figure S8). Thus, the Barker catalogue does not independently replicate the Rayleigh phase-clustering result, but it supports the weaker claim that both glacial background state and precession phase carry conditional predictive information in an event catalogue independent of the Rousseau speleothem detections.

Discussion

The main result is that DO-like Asian monsoon transition probability contains two separable layers of predictive information. The event-process baseline controls recent same-type event occurrence and local composite sampling resolution, so the subsequent likelihood gains are not simply consequences of event clustering or better resolved portions of the speleothem composite. Conditioned on the event-process baseline, LR04+CO₂ improves event-rate prediction for both transition directions. This is consistent with the broad view that MCV depends on background climate state (*Barker et al.*, 2011; *Zhang et al.*, 2017; *Vettoretti et al.*, 2022; *Malmierca-Vallet and Sime*, 2023), but the contribution here is quantitative and event based: over 0–640 kyr, slow glacial background predictors improve the likelihood of the observed DO-like monsoon transition sequence after the event-process baseline is included. Precession phase then adds a second layer of information after LR04 and CO₂ are included.

The lagged predictive-information scan clarifies how these two layers differ in time. For strong starts, the best single-driver lags are 2.4 kyr for LR04, 1.8 kyr for CO₂, and 0 kyr for precession phase (Figure 4); for weak starts, the climate-state lags are shorter, with 0 kyr for LR04, 0.8 kyr for CO₂, and 0 kyr for precession phase. LR04 and CO₂ therefore behave as slowly varying background-state predictors that index ice volume, deep-ocean temperature, greenhouse forcing, and boundary conditions relevant to AMOC and sea-ice–ocean stability (*Zhang et al.*, 2017; *Vettoretti et al.*, 2022; *Zhang et al.*, 2021). This event-rate perspective is consistent with *Lohmann and Ditlevsen* (2018), who found that last-glacial DO residence times improved when ice volume and insolation modulated transition statistics; here we reach a related inference over the full 640 kyr, with event history and sampling resolution explicitly controlled. Precession phase, by contrast, indexes the seasonal geometry of low-latitude insolation forcing (*Berger*, 1978; *Berger and Loutre*, 1991). Its near-zero best lag is consistent with an orbital-seasonal transition-window variable rather than a slow integrator like ice volume or greenhouse gases.

The strong–weak lag contrast in the climate-state predictors should be treated as a more tentative, direction-specific feature of the result. Generic hysteresis can justify path dependence and different transition thresholds (*Stommel*, 1961; *Ganopolski and Rahmstorf*, 2001), but it does not by itself predict why the climate-state lag should be longer for strong starts than for weak starts. The empirical pattern instead suggests that the two transition directions sample the slow background state differently: strong-monsoon, interstadial-like starts retain more predictive information from earlier LR04/CO₂ conditions, whereas weak-monsoon starts are more closely aligned with the contemporaneous background state. This direction-dependent behavior is consistent with statistical analyses showing that stadial and interstadial residence times need not respond to the same controls (*Lohmann and Ditlevsen*, 2018), and with model studies in which DO-like variability occurs within restricted background-state regimes rather than as a simple linear response to one forcing (*Zhang et al.*, 2017; *Vettoretti et al.*, 2022). Asian speleothem records integrate both background-state and orbital-seasonal forcing (*Cheng et al.*, 2016; *Corrick et al.*, 2020; *Xie and Liu*, 2023), so the data are

consistent with a background-susceptibility plus orbital-phase-organization view, not deterministic astronomical triggering of individual transitions.

The opposite preferred sectors for strong and weak starts imply that precession also influences transition direction, although this need not map onto a single model’s phase curve. A more robust link to *Zhang et al.* (2021) is dynamical: in their COSMOS experiments, precession-driven seasonal-insolation changes alter tropical North Atlantic hydroclimate and salinity under intermediate glacial boundary conditions, helping shift the AMOC into a DO-like oscillatory regime via subpolar sea-ice–ocean feedbacks. Our proxy result extends this picture with an event-timing constraint: after accounting for LR04 and CO₂, the probability of entering versus leaving a strong-monsoon state is distributed asymmetrically across the precession cycle. We interpret this as evidence for precession-modulated transition windows—direction-dependent and asymmetric—rather than a precise reproduction of any specific model’s phase response or exact antiphased astronomical pacing.

The absence of comparable obliquity or 65°N summer-solstice-insolation contributions should be read narrowly. These forcings may still affect MCV amplitude or background DO-like stability (*Sun et al.*, 2021; *Hodell et al.*, 2023; *Zhang et al.*, 2021; *Malmierca-Vallet and Sime*, 2023); they simply do not add a separable event-timing signal in this predictive model after event history, sampling resolution, LR04, CO₂, and precession phase are included. The weak AICc improvement from 65°N insolation for weak starts is therefore a sensitivity flag, not an alternative to the phase result.

Several limitations counsel caution. First, the event catalogue depends on the Rousseau et al. KS detection procedure applied to the Cheng et al. speleothem composite. Because the composite contains strong precession-band variability, the KS algorithm could in principle detect high-gradient portions of the orbital waveform rather than only abrupt millennial transitions, which would overestimate the strength of precession-phase organization. We cannot fully exclude this. However, the Barker et al. catalogue—independent of both the Cheng composite and Rousseau detections—still yields significant conditional precession information, supporting a real (if weaker) association. We therefore regard the event-timing signal as unlikely to be purely a KS artifact, while acknowledging that the main speleothem analysis may overestimate its magnitude.

Second, the speleothem catalogue is an indirect proxy for MCV rather than a complete record of the North Atlantic phenomenon itself. Chinese speleothem $\delta^{18}\text{O}$ records abrupt monsoon-system responses that are often tightly linked to Greenland stadial–interstadial changes, but the signal also integrates upstream rainout, moisture-source changes, circulation pathways, seasonality, and regional hydroclimate (*Liu et al.*, 2014; *Cheng et al.*, 2021; *Zheng et al.*, 2022). The “strong” and “weak” starts analyzed here should therefore be interpreted as speleothem-recorded DO-like Asian monsoon transitions, not as a one-to-one catalogue of all MCV events.

Third, the binned Poisson predictive-information framework is a statistical approximation. Sparse-bin diagnostics show no overdispersion, bootstrap LR tests support the main comparisons (Text S5), and bin-width sensitivity confirms qualitative stability up to 1 kyr. Nonetheless, the likelihood is conditional on the included history and covariate terms, does not resolve event-duration dynamics, and should not be read as a mechanistic point-process model for the full climate system. The likelihood-ratio tests establish statistical association and predictive improvement, not a complete dynamical mechanism. Fourth, predictors such as precession phase, insolation, eccentricity, and ice-volume indices are not independent in the climate system; our sensitivity results show that additional variables do not improve prediction in this framework, but this should not exclude indirect pathways through which obliquity, eccentricity, or ice sheets influence MCV.

The information gains are statistically robust but modest—several to ten percent improvement in per-event likelihood—consistent with a stochastic threshold-crossing component alongside the

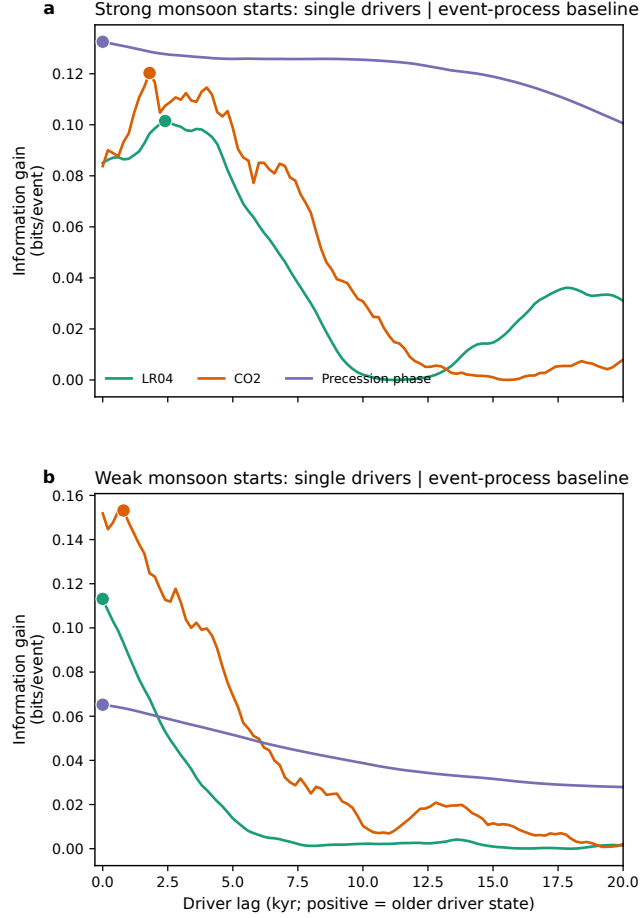


Figure 4: Lagged predictive information from individual drivers. Model-based predictive-information gain, expressed as bits per event, from adding lagged LR04 (*Lisiecki and Raymo, 2005*), CO₂ (*Bereiter et al., 2015*), or precession phase (*Laskar et al., 2004*) to the event-process baseline containing same-type event history and sampling resolution. (a) Strong monsoon starts. (b) Weak monsoon starts. Positive lags indicate that the driver is sampled from an older climate state relative to the event bin.

systematic phase and background-state organization. Together, the Rayleigh tests, the two-layer predictive-information hierarchy, and the sensitivity experiments point to coupled background-state susceptibility and precession-phase modulation of abrupt monsoon-transition probability.

Conclusions

This study indicates that DO-like Asian monsoon transition timing in the 0–640 kyr BP Chinese speleothem composite is organized by both slow glacial background state and precession phase. After event history and sampling resolution are controlled, LR04+CO₂ improves event-rate prediction, and precession phase adds further information beyond this background state. Strong and weak monsoon starts also prefer opposite sectors of the precession cycle, suggesting direction-dependent phase organization rather than a simple increase in overall millennial variability. Lagged diagnostics support a temporal distinction between slower LR04/CO₂ background control and near-contemporaneous precession-phase information. These results are stable across bin-width, KS-window, and age-uncertainty sensitivity tests, and are not replaced by obliquity, eccentricity, Antarctic temperature, or 65°N summer insolation as independent predictors. The event-timing perspective therefore extends amplitude-based studies of orbital–MCV interactions and provides a probabilistic target for transient climate-model experiments on how orbital forcing and glacial boundary conditions combine to shape abrupt hydroclimate transitions.

Data Availability

All data used in this study are from previously published sources: the Chinese speleothem composite (*Cheng et al.*, 2016), the 196 abrupt transition catalogue (*Rousseau et al.*, 2023), the Barker et al. D-O warming catalogue used for the independent sensitivity check (*Barker et al.*, 2011), LR04 benthic $\delta^{18}\text{O}$ stack (*Lisiecki and Raymo*, 2005), atmospheric CO₂ from EPICA Dome C (*Bereiter et al.*, 2015), EDC δD (*Jouzel et al.*, 2007), and orbital solutions (*Laskar et al.*, 2004). The analysis code is available at https://github.com/Peisong-Zheng/DO_warming.

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Competing Interests

The authors declare no competing interests.

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